THE UNIVERSITY OF HONG KONG
DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE

STAT7008 Programming for Data Science

Application of Python Programming on US Stock Market Investment

Note: This is a group project and the maximum number of students in a group is four. Once you have formed a group, you need to elect a group leader, who will be responsible for submitting the investment decision, written report and presentation video on behalf of your group. The group leader needs to fill in the Group Information Excel file on Moodle (under Project Section) and submit the file to Group Information submission link (also under Project Section) before September 13. If you have troubles finding teammates, please contact me on hcheung4@hku.hk as soon as possible.

Project Description: This is a Python programming project involving three investment exercises, i.e., daily, weekly and monthly investments. For each investment exercise, you need to imagine that you have been given 1 million virtual US dollars and you need to invest all the money in (at most) 5 companies listed in the US stock market based on the closing prices of the stocks on November 6. Therefore, you will invest 3 million virtual US dollars for the entire project. Assume there is no transaction cost, and you cannot borrow money or short sell. You are not allowed to invest in ETFs (exchange-traded funds) or indices. Your daily, weekly and monthly investments will be evaluated based on the closing prices of the chosen stocks on November 9, November 13, and December 4, respectively. The chosen stocks for the three investment exercises may or may not be the same. We use closing price (not adjusted closing price) in this project. You should notify me and the marking tutor (Jiacheng) if there has been a dividend payment (i.e., ex-dividend date) during the evaluation periods.

You may use what you have learned from this course or other HKU courses to find the stocks that could maximize your portfolio return. Additionally, any statistical, econometric, or financial methods learned through library research (or from the internet) may also be used. Throughout the semester, I will discuss how I would approach this project if I was a student in this course. But you don't have to follow my ideas. In fact, I would be much happier if you could present your own ideas in the written report.

There are two deadlines for this project. The first is November 8 (23:59) when the leader submits the investment decisions for all three exercises. The second is December 6 (23:59) when the leader submits the written report (in pdf format), Python code and sample datasets (in a single zip folder), and presentation video (in mp4 format). You are strongly encouraged to use Zoom to record the presentation video (https://its.hku.hk/software/zoom/).

The contributions of the written report, computer code, presentation video and portfolio performance to the final grade of this project are 60%, 5%, 20% and 15%, respectively. The written report must be at least two pages long and it must be written by humans. Make sure you write every single sentence by yourself, and there is no need to ask AI for help with your grammar. You can use AI agent(s) to write Python code, and you only need to submit some small sample datasets (if you have used too many datasets to be submitted). The presentation video must be at most 20 minutes long, and each student must present in the video. For example, if the presentation lasts for 20 minutes and there are four students in your group, then each student should present for five minutes. Your portfolio

performance will be graded based on the performance ranking across all groups. Having said that, I will definitely give you some marks for your portfolio performance even if your group is ranked at the bottom of the list. You will be given an individual grade for the presentation and the grades for the written report, coding and portfolio performance will be shared by all students in the group.